

Calibrated to Whom? Persona and Language Effects on Cultural Values in JEV

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Abstract

Decision-only language models return a probability for every answer option instead of generating text, which makes them attractive as survey respondents and as judges. We audit the cultural values of one such model, TypeSafe’s JEV, with the Values Survey Module 2013. We asked it the 24 items as 12 matched Saudi and 12 matched American personas and without a persona, in English and Arabic, under eight ways of formulating the request (288,000 answers). JEV’s answers were highly repeatable (ICC 0.997), and without a persona they resembled those of its own American personas. When the persona was Saudi rather than American, the answers moved in the direction of the human Saudi–US difference, reproducing 87% of its size in English but 62% in Arabic, with long-term orientation reversed. A language cross shows that the smaller difference in Arabic comes from the language of the items, not from the language of the persona description. Age shifted the profiles about as much as nationality, gender shifted them more for Saudi than for American personas, and JEV was less confident in Arabic and for Saudi personas. These patterns held in every request design, although the model never generates text.

1 Introduction

Value surveys have become a standard way to ask whose values a language model reflects (Santurkar et al., 2023; Durmus et al., 2023; Cao et al., 2023; AlKhamissi et al., 2024; Tao et al., 2024). With generative models the procedure is indirect: many completions are sampled, free text is mapped back to answer options, refusals have to be handled, and the results move with the format of the prompt (Röttger et al., 2024).

A new class of models removes these steps. JEV, released by TypeSafe AI in September 2026, is described as a “System One” model: given a context and a question with fixed options, it returns a probability for every option and never generates text

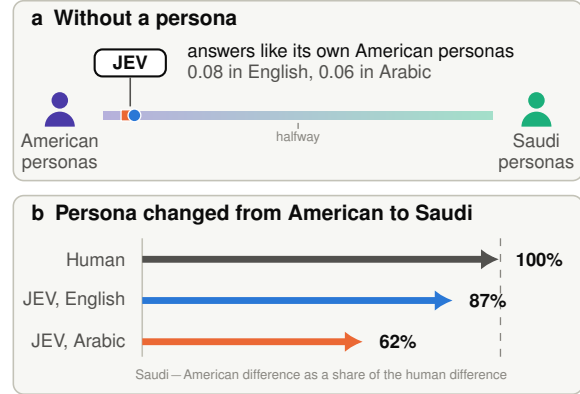


Figure 1: Calibrated to whom? (a) Without a persona, JEV’s answers lie next to those of its own American personas, on a scale from its American (0) to its Saudi (1) personas. (b) When the persona changes from American to Saudi, the answers move by 87% of the Saudi–US difference in Hofstede’s scores in English and by 62% in Arabic (documented set-up).

(TypeSafe AI, 2026). Within weeks it was evaluated as an inexpensive first-pass judge (Li et al., 2026) and used to predict the answers of described respondents in population experiments (Li, 2026). Both studies worked in English, and neither asked whose values the model itself reports.

For a model that stands in for respondents, three questions matter. What does it answer when no respondent is described? When the respondent’s nationality is given, do its answers move in the direction of real national differences, and how far? And do the answers depend on the language of the questionnaire? Figure 1 previews our answers. The language can matter even for a single item: asked in English whether an organisation’s rules should never be broken, JEV’s Saudi personas agree more strongly than its American personas, whereas in Arabic the two answer alike. Arabic makes the last question pressing: cultural biases in Arabic contexts are well documented for generative models (Naous et al., 2024; AlKhamissi et al., 2024), and

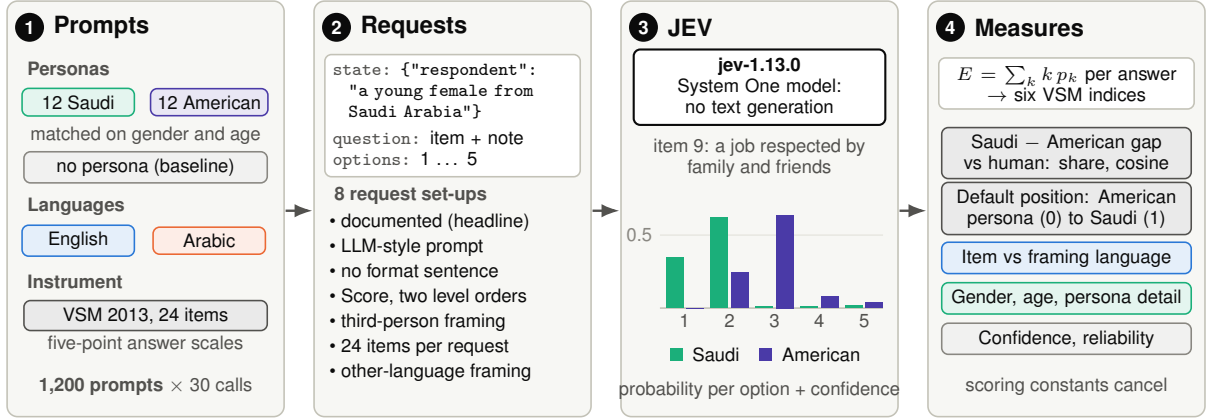


Figure 2: Study design. (1) Matched Saudi and American personas and a no-persona baseline, in English and Arabic, answer the 24 VSM 2013 items. (2) Each prompt becomes a JEV request in eight set-ups. (3) JEV returns a probability for each answer option; the bars show its mean probabilities for item 9 (“how important would it be to you to have a job respected by your family and friends”) for a young Saudi and a young American woman (documented set-up, English; 1 = of utmost importance). (4) Expected answers are scored on the six VSM dimensions and compared through measures in which the scoring constants cancel.

JEV is trained mainly on English, with lower documented accuracy in other languages (TypeSafe AI, 2026).

We answer these questions with the Values Survey Module 2013 (VSM 2013; Hofstede and Minkov, 2013b). We describe 12 matched Saudi and 12 matched American personas, or none, in English and in Arabic, and compare the Saudi–American difference in JEV’s answers with the human difference between the two countries (Figure 2). Because the absolute position of VSM scores depends on constants that are not published for the reference countries (Bulté and Rigouts Terryn, 2026), every measure compares two of JEV’s own profiles, so that the constants cancel. Each of the 1,200 prompts was sent 30 times under eight ways of formulating the request, 288,000 answers in total.

Our contributions are:

- the first audit of the cultural values of a decision-only model, in English and Arabic, with constant-free measures of the national difference and of the default profile;
- evidence that JEV’s answers follow the human Saudi–US difference in direction and in 87% of its size in English but 62% in Arabic, that the language of the items rather than of the persona description carries this gap, and that age and gender shift the profiles in stereotyped ways;
- a robustness analysis over eight request de-

signs that separates findings that depend on how the model is asked from those that do not, including the effect of errors in a translated answer scale.

2 Related Work

Survey-based cultural evaluation. Studies of cultural values in language models administer value surveys and compare the answers with national results, using Hofstede’s dimensions (Cao et al., 2023; Tao et al., 2024; Masoud et al., 2025) or the World Values Survey and related opinion polls (Santurkar et al., 2023; Durmus et al., 2023; AlKhamissi et al., 2024). Recurring findings are that answers resemble those of the United States or other Western countries (Cao et al., 2023; Durmus et al., 2023; Tao et al., 2024), also when the question is asked in another language (Wang et al., 2024); that naming a country or using its language moves the answers only partway towards it (Durmus et al., 2023; AlKhamissi et al., 2024; Tao et al., 2024); and that answers given for a group can reproduce stereotypes (Durmus et al., 2023). In Arabic contexts, models favour Western entities and norms (Naous et al., 2024), and GPT-4’s answers to the VSM 2013 matched Chinese values better than American or Arab ones (Masoud et al., 2025). We use the same family of instruments, but for a model that does not generate text.

Validity of survey measurement. Forced-choice answers from language models can differ from their unconstrained responses and change with small

variations of the prompt (Röttger et al., 2024). For the VSM, each dimension index contains a constant chosen by the researcher, so the distance between a model’s scores and published country scores is not identified (Bulté and Rigouts Terryn, 2026). Lacking these constants, Masoud et al. (2025) compared the order of countries on each dimension with the order of their published scores instead. We therefore compare profiles measured on the same scale and vary the request design systematically.

Language models as respondents. Conditioning a model on demographic descriptions has been proposed as a way to simulate survey samples (Argyle et al., 2023), while other work shows that model opinions match some groups better than others and that steering towards a group succeeds only in part (Santurkar et al., 2023). Pluralistic alignment methods aim to represent several cultures at once (Yuan et al., 2024).

JEV. JEV is a proprietary model released by TypeSafe AI in September 2026 as the first of what the company calls “System One” models, named after the fast, intuitive mode of thinking described by Kahneman (2011) (TypeSafe AI, 2026). It does not generate text. Given a context, the *state*, it answers typed questions: a Choice question returns a probability for each of up to 255 named options, a Score question a probability for each of two to ten ordered levels, and a yes/no question the probability of yes. All probabilities are produced in a single non-autoregressive pass, and each answer carries a confidence value. TypeSafe trains the model with what it calls reinforcement learning for calibrated decisions, positions it as a fast and inexpensive decision component for software, and notes that it is trained mainly on English and is less accurate in other languages. Li et al. (2026) found it within three points of a frontier judge on preference and factuality judgements, and weaker where a judgement requires checking a derivation, at a fraction of a percent of the cost. Li (2026) used it to predict how described respondents would answer in population experiments, with English text only. Neither study examined cultural values or Arabic.

3 Experimental Setup

3.1 The Model

Typed questions and answers. We used two of JEV’s question types. A *Choice* question lists named options with descriptions; its answer gives

the most probable option, a probability for every option and a confidence value between 0 and 1 derived from the shape of the distribution. A *Score* question lists ordered levels and returns a probability for each. The state can be text or a JSON object, questions in one request are answered independently, and the model has no temperature or seed parameter (TypeSafe AI, 2026).

Access. We sent all requests to TypeSafe’s API and asked explicitly for version jev-1.13.0, which every response reported. The data were collected on 28 and 29 September 2026. Probabilities are returned with two decimals; we renormalised them to sum to one. Cost and software are listed in Appendix G.

3.2 Instrument and Reference Values

VSM 2013. We used the 24 content items of the VSM 2013, each answered on a five-point scale, and computed the six dimension indices (power distance, individualism, masculinity, uncertainty avoidance, long-term orientation, indulgence) with the manual’s formulas (Appendix A).

Human references. We used Hofstede’s scores for Saudi Arabia and the United States (Hofstede et al., 2010) and, as an alternative Saudi profile, the VSM 2013 scores of Almutairi et al. (2021). With Hofstede’s scores, the human Saudi-minus-US difference h is a vector of length 94.3 (Appendix A).

Constant-free comparisons. Each index contains an additive constant chosen by the researcher, and the constants behind published country scores are not reported, so the absolute distance between a model’s indices and a country’s scores is not identified (Bulté and Rigouts Terryn, 2026). All our measures compare two profiles on the same scale, so the constants cancel.

3.3 Personas, Prompts and Languages

Matched personas. For each country, a persona names the country alone, the country with a gender (female, male) or an age group (young, middle-aged, elder), or the country with both. The resulting 12 demographic profiles are the same for Saudi Arabia and the United States, so Saudi and American personas can be compared in matched pairs. A baseline asks the items without a persona.

Prompts. English items follow the official VSM 2013 wording, recast as questions (“In choosing an ideal job, how important would it be to you to have

sufficient time for your personal or home life?”); Arabic items follow the official Arabic translation (Hofstede and Minkov, 2013a), recast in the same way. A check against the official Arabic questionnaire found three deviations in the answer options and opening line of items 15 to 20 in our first runs; we corrected them and asked these items again in every set-up (Appendix C).

Repetitions. With 25 persona conditions, two languages and 24 items there are 1,200 prompts. Each was sent 30 times in each of the eight set-ups, giving 36,000 answers per set-up.

3.4 Request Set-ups

JEV has no system and user roles, so a persona prompt can be passed to it in several ways (Table 1; texts in Appendix B).

Documented set-up. Our headline set-up follows TypeSafe’s recommendation to give content as named fields of the state and the task in the question. The state is $\{\text{"respondent": "a young female from Saudi Arabia"}\}$, and a Choice question contains the item wording with the note “Answer as the respondent described in ‘respondent’ would answer” and the five answer labels as options.

LLM-style prompt and variations. Set-up 2 passes a conventional persona prompt unchanged: the system prompt (“Imagine you are a young female from Saudi Arabia. Answer the following question from your perspective as a person from this cultural background.”), followed by a format sentence) as the state, and the user prompt, with the numbered options and the item, as the question. Set-ups 3 to 7 each change one element: they drop the format sentence, use a Score question with the levels in questionnaire or reverse order, describe the respondent in the third person and ask which answer they would choose, as in population experiments with JEV (Li, 2026), or send the 24 items of a persona as 24 questions in one request.

Language cross. Set-up 8 takes the system prompt from the other language, so that English framing meets Arabic items and Arabic framing meets English items.

3.5 Measures

Answers and profiles. For each call we computed the expected answer $E = \sum_k k p_k$ from the probabilities $p_1, \dots, p_5$, averaged E over the 30

	Set-up	Request
1	Documented (headline)	Persona as a field of the state; item wording and a note; Choice
2	LLM-style prompt	System prompt as state; user prompt as question; Choice
3	No format sentence	As 2, without “IMPORTANT: Respond with ONLY ...”
4	Score question	As 2, options as ordered levels
5	Score, reversed	As 4, levels in reverse order
6	Third person	“Description of the survey respondent: [persona].”; “Which answer would they choose?”
7	24 items per call	As 2, 24 questions per request
8	Other-language framing	As 2, system prompt of the other language

Table 1: The eight request set-ups. Each covers the 1,200 prompts, each sent 30 times.

calls of each prompt and computed the six indices for every persona condition and language.

National difference. For each of the 12 demographic profiles i and each language, $\mathbf{d}_i$ is the index vector of the Saudi persona minus that of the American persona. From the mean $\bar{\mathbf{d}}$ we computed the share of the human difference reproduced along its direction and the direction cosine,

$$s = \frac{\bar{\mathbf{d}} \cdot \mathbf{h}}{\|\mathbf{h}\|^2}, \quad \cos = \frac{\bar{\mathbf{d}} \cdot \mathbf{h}}{\|\bar{\mathbf{d}}\| \|\mathbf{h}\|}.$$

A share of 100% is a difference of the human size in the human direction.

Default position. The no-persona profile $\mathbf{b}$ is placed between JEV’s own American and Saudi personas, with mean profiles $\bar{\mathbf{u}}$ and $\bar{\mathbf{s}}$, by $t = (\mathbf{b} - \bar{\mathbf{u}}) \cdot (\bar{\mathbf{s}} - \bar{\mathbf{u}}) / \|\bar{\mathbf{s}} - \bar{\mathbf{u}}\|^2$: 0 means that the default resembles the American personas, 1 the Saudi personas.

Gender, age and detail. Female-minus-male differences compare personas matched on country, language and age (four pairs per country and language); elder-minus-young differences are matched on gender (three pairs). The share s is also computed separately for country-only personas and for personas with one or two further attributes.

Confidence and reliability. Confidence is averaged over the 30 calls of each prompt and compared across matched prompts. Test-retest reliability is the one-way intraclass correlation of E across the 30 calls.

Rank agreement. We also applied the rank test of Masoud et al. (2025), which asks whether a model orders countries on each dimension as their human scores do, using Kendall’s τ and the share of dimensions out of order. With two countries, the order on a dimension is the sign of the Saudi-minus-US difference, so τ is +1 or −1 for each dimension; we report the number of dimensions in the human order, k , and the mean $\tau = (2k - 6)/6$. As in their model-level comparison, we also let the language stand for the country and compared the no-persona profile in Arabic with that in English. The constants do not affect the order of two countries on a dimension.

3.6 Statistical Analysis

Repeated calls return nearly identical answers and are not independent observations, so matched personas are the units of replication. Intervals for dimension differences are 95% t -intervals across matched profiles or pairs; intervals for shares and their differences are percentile intervals from 4,000 bootstrap resamples of the 12 matched profiles. Confidence differences are tested with Wilcoxon signed-rank tests over matched prompts. The analysis is exploratory and was not pre-registered; as a safeguard, every finding was re-estimated in all eight set-ups.

4 Results

Unless stated otherwise, results are for the documented set-up.

4.1 Reliability

JEV’s answers were highly repeatable. The intra-class correlation of the expected answer over 30 calls was 0.997 in the documented set-up and between 0.995 and 0.998 in the others, the standard deviation across calls averaged 0.03 on the five-point scale, and the most probable option was the same in all 30 calls for 83–89% of prompts (Appendix D). Mean confidence ranged from 0.49 to 0.58 across set-ups, with the flattest distributions for Score questions.

4.2 The Default without a Persona

Without a persona, JEV answered like its own American personas: the default profile lay at 0.08 in English and 0.06 in Arabic on the scale from its American (0) to its Saudi (1) personas. In 15 of the 16 combinations of set-up and language the position lay between −0.26 and 0.13 (Appendix E);

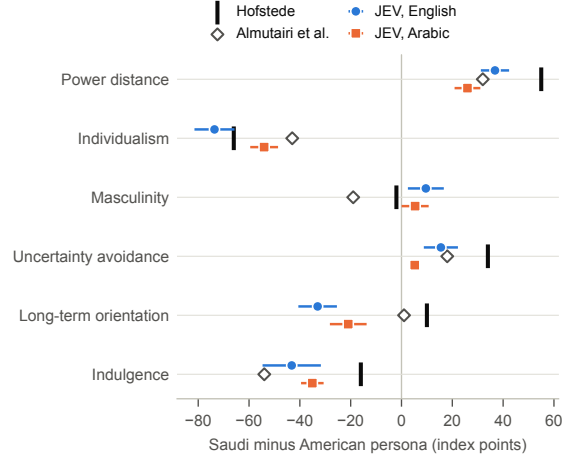


Figure 3: Saudi minus American persona on the six VSM dimensions (documented set-up; means over the 12 matched profiles with 95% CIs), with the human differences from Hofstede’s scores and from Almutairi et al. (2021).

with the LLM-style prompt, the English default lay beyond the American personas (−0.25). The exception was the third-person framing in Arabic, which placed an undescribed respondent midway between the two nationalities (0.53).

4.3 The Saudi–American Difference

When the persona changed from American to Saudi, JEV’s answers moved in the human direction (Figure 3). In English, its Saudi–American difference amounted to 87.1% of the Hofstede difference (95% CI 80.5–93.9), with a direction cosine of 0.82; against the Saudi profile of Almutairi et al. (2021) the share was 104.1% (94.8–113.6).

By dimension. Four of the five dimensions on which the two countries differ came out in the human direction: Saudi personas scored higher on power distance (+36.8; Hofstede +55) and uncertainty avoidance (+15.5; +34) and lower on individualism (−73.6; −66) and indulgence (−43.2; −16). Masculinity, on which the countries barely differ (−2), showed a small positive difference (+9.6).

Long-term orientation. Saudi personas scored 33.0 points lower on long-term orientation (−40.6 to −25.4), whereas Hofstede’s Saudi profile is 10 points higher. The VSM index counts greater importance of doing a service to a friend and greater national pride towards short-term orientation, and JEV’s Saudi personas gave both: on the 1–5 answer scale, their answers were 0.48 points lower (more

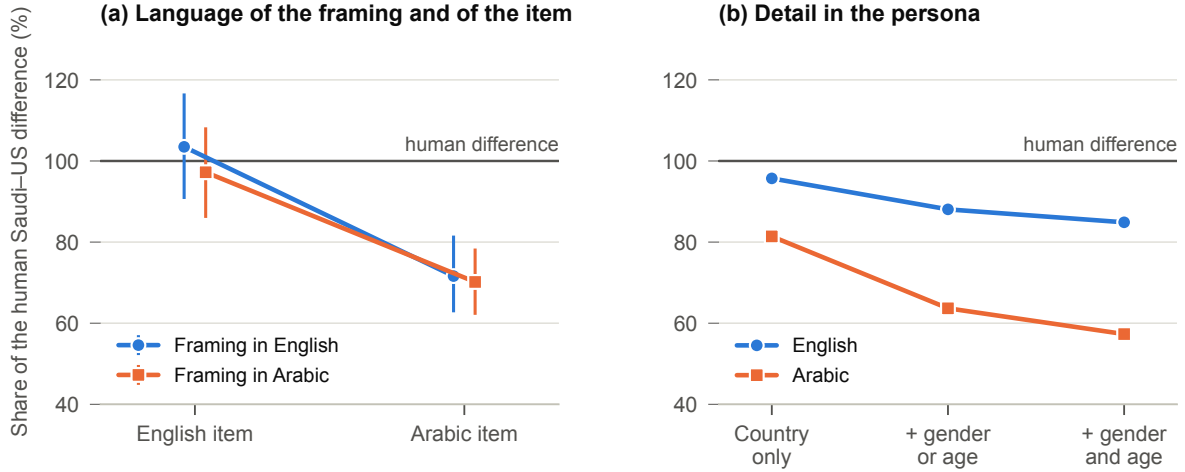


Figure 4: Share of the human Saudi-US difference reproduced. (a) By the language of the framing and of the item (LLM-style prompt and its language cross; 95% bootstrap intervals over matched profiles). (b) By the detail in the persona (documented set-up): country only (one matched pair per language), one further attribute (five pairs) and two (six pairs).

important) on the service item in both languages and 0.77 (English) and 0.84 (Arabic) points lower (prouder) on national pride.

4.4 Language

In Arabic, JEV reproduced 62.0% of the Hofstede difference (56.3–67.9; cosine 0.80), 25.1 points less than in English (–31.0 to –20.6). The direction held on every dimension, but the gap narrowed on power distance (+26.0), individualism (–54.1) and uncertainty avoidance (+5.2).

Item language, not framing. The language cross separated the two sources (Figure 4a; LLM-style prompt). With English items, the share was 103.5% under English framing and 97.2% under Arabic framing; with Arabic items, it was 71.6% and 70.2%. Averaged over framings, an Arabic item lowered the share by 29.5 points (26.1–33.1), whereas an Arabic framing changed it by –3.9 points (–8.9 to 1.5).

By item. The national difference disappeared in Arabic for two agree–disagree statements, on avoiding two bosses and on never breaking an organisation’s rules (English –0.30 and –0.54 on the answer scale; Arabic 0.00 and –0.06; Appendix F).

Persona detail. More detail in the persona weakened the national difference in both languages (Figure 4b): in English from 95.7% for country-only personas to 88.1% with gender or age and 84.9% with both; in Arabic from 81.4% to 63.7% and

57.3%.

4.5 Gender and Age

Female Saudi personas were more individualist (+19.0, 15.1–22.9; 8 of 8 matched pairs), less masculine (–16.7; 8 of 8) and less indulgent (–20.2; 7 of 8) than male Saudi personas (Figure 5). Among American personas the gender pattern was weaker (individualism +8.9, masculinity –11.3) and absent for indulgence (–1.6). Age changed the profiles more, and alike for both nationalities: elder Saudi personas scored higher than young ones on power distance (+27.4) and uncertainty avoidance (+32.9) and lower on individualism (–57.6), masculinity (–87.7) and indulgence (–49.6), each in 6 of 6 pairs, with similar values for American personas. The age gap in individualism was as large as the national gap (–73.6 in English, –54.1 in Arabic), and the age gap in masculinity exceeded any national difference.

4.6 Confidence

JEV was less confident in Arabic, by 0.090 (English higher in 72% of 600 matched prompts; Wilcoxon $p = 3.3 \times 10^{-33}$), and for Saudi personas, by 0.024 (American higher in 58% of 576 matched prompts; $p = 4.4 \times 10^{-5}$). The gap between American and Saudi personas appeared in all eight set-ups and the gap between English and Arabic in seven. The exception, the language cross, gave higher confidence to Arabic items under English framing than to English items under Arabic

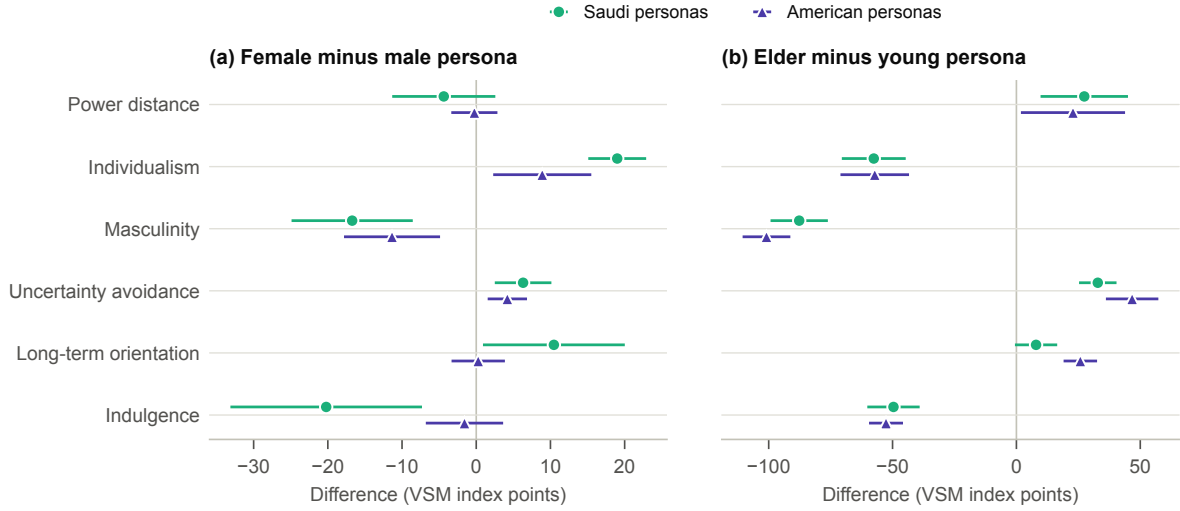


Figure 5: Gender (a) and age (b) differences within Saudi and American personas (documented set-up; means over matched pairs in both languages, 8 gender and 6 age pairs per nationality, with 95% CIs). The two panels use different scales.

framing: confidence followed the language of the framing, whereas the national difference followed the language of the items. Confidence was lowest for the agree–disagree items on managers (0.35) and on rules (0.38) and highest for how often one feels nervous or tense (0.75).

4.7 Robustness to Request Design

The set-up changed the size of the effects but not their direction (Figure 6; Appendix E). The English share ranged from 87.1% to 106.8% and the Arabic share from 42.1% (third person) to 72.2%. In every set-up the Arabic share was smaller than the English share, the English difference pointed in the human direction (cosine 0.82–0.85), female Saudi personas were more individualist and less masculine than male ones, elder Saudi personas scored higher on power distance and lower on individualism than young ones, and confidence was lower for Saudi personas. Sending the 24 items in one request left the results practically unchanged (English 103.8% against 103.5%, Arabic 69.9% against 70.2%). The Arabic answer options mattered as well: with the deviating options of our first runs, the Arabic share was 42.7% instead of 62.0%, and the power-distance difference had the wrong sign (−10.5 instead of +26.0; Appendix C).

4.8 Rank Agreement

In the rank test of Masoud et al. (2025), JEV ordered Saudi Arabia and the United States as Hofstede’s scores do on four of the six dimensions

in both languages: power distance, individualism, uncertainty avoidance and indulgence, but not masculinity or long-term orientation (mean $\tau = 0.33$; 33% of dimensions mis-ranked). The profile of Almutairi et al. (2021) gives the same result, as its differences have the same signs. Across set-ups, four or five dimensions were in order in English and four in Arabic (Figure 6; Table 8). Long-term orientation was out of order in every set-up and language, and masculinity, on which Hofstede’s scores differ by 2 points, in all eight set-ups in Arabic and in five in English. No matched profile had all six dimensions in order in any set-up. Without a persona, the language alone did not produce the national order: the Arabic profile differed from the English one in the human direction on one dimension in the documented set-up (individualism; $\tau = -0.67$) and on three or four in the others.

5 Discussion

5.1 Cultural Patterns without Text Generation

JEV shows patterns that have been reported for generative models: an American default, movement towards a named country, a weaker national difference in a language other than English, and stereotyped answers for demographic groups (Durmus et al., 2023; Tao et al., 2024; AlKhamissi et al., 2024; Wang et al., 2024). In JEV these patterns arise without decoding, sampling or refusals. This suggests that they reside in what the model has

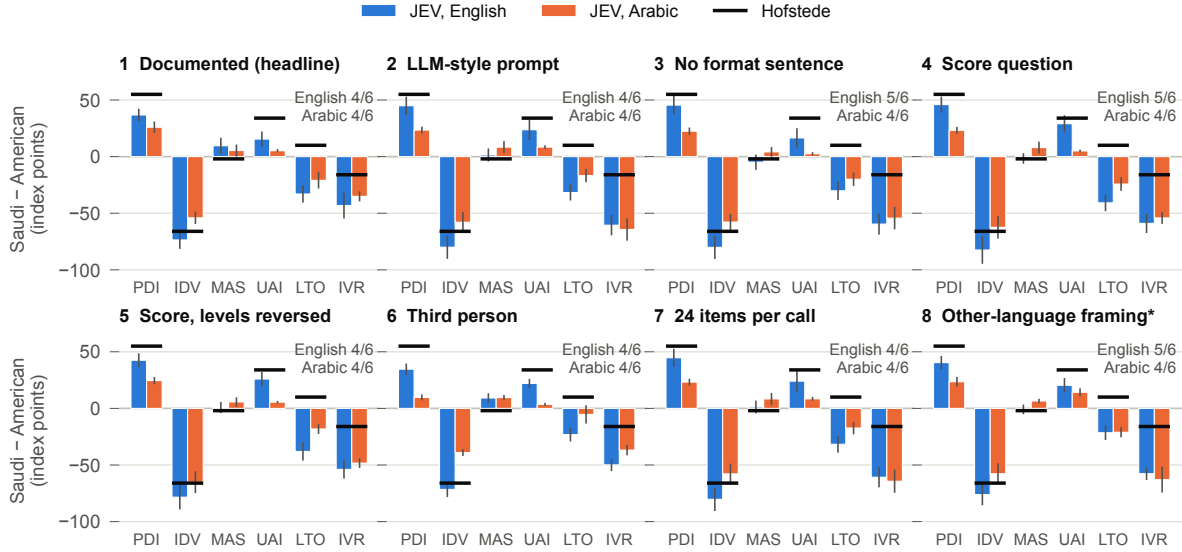


Figure 6: Rank test by set-up. Bars: Saudi minus American persona on each dimension (means over the 12 matched profiles with 95% CIs); black lines: the Hofstede difference. A bar on the other side of zero from its line marks a dimension on which JEV orders the two countries differently from Hofstede’s scores. Top right: dimensions in Hofstede’s order, of six. PDI: power distance, IDV: individualism, MAS: masculinity, UAI: uncertainty avoidance, LTO: long-term orientation, IVR: indulgence. *English and Arabic refer to the items; the framing is in the other language.

learned about groups, languages and survey items rather than in the generation of text, and that replacing generation with calibrated decisions does not by itself remove them.

5.2 Language: the Question, Not the Persona

The language cross places the Arabic effect in the item wording (−29.5 points) rather than in the language of the persona description (−3.9). The largest item-level losses were in agree–disagree statements about organisational hierarchy and rules, which suggests that the Arabic formulations of these statements carry less of the national contrast than their English counterparts. For multilingual evaluation this means that the translation of the items, not only the language of the prompt, is part of what is measured. Nor did the language stand in for the culture: without a persona, Arabic items did not move JEV towards the Saudi order, a caution for comparisons that let the prompt language represent a country (Masoud et al., 2025). Our first runs illustrate the stakes: deviating answer options in three Arabic items reversed the sign of the power-distance difference.

5.3 Personas as Population Proxies

Age shifted JEV’s profiles about as much as nationality, gender shifted them in stereotyped directions,

and adding either attribute diluted the national difference (in English from 96% to 85%, in Arabic from 81% to 57%). Population experiments that condition JEV on several demographic attributes (Li, 2026) should therefore expect the national signal to weaken as attributes are added, and age to carry stereotyped contrasts, such as a masculinity gap between elder and young personas larger than any national difference. The larger gender contrast for Saudi than for American personas is a further stereotype that simulated Saudi samples would inherit.

5.4 Confidence as a Signal

JEV was less confident in Arabic and for Saudi personas, and its confidence followed the language of the framing rather than of the items. Lower confidence in these contexts could help to flag answers that rest on less evidence. It does not show that the model is calibrated to the diversity of real respondents: a value question has no correct option, and confidence measures the model’s certainty, not the agreement among people. Testing whether JEV’s probabilities match the answer distributions of Saudi respondents, for example in the World Values Survey or the Arab Barometer, is the natural next step.

6 Conclusion

We audited the cultural values of a decision-only model with the VSM 2013, matched Saudi and American personas, two languages and eight request designs. JEV’s answers are highly repeatable, American by default and responsive to nationality: in English they reproduce most of the human Saudi–US difference in its direction, and in Arabic less, because of the item wording rather than the language of the persona. Age and gender shift the profiles in stereotyped ways, and confidence is lower in Arabic and for Saudi personas. These findings hold across request designs. A model that never writes text is thus not free of the cultural patterns known from generative models, and using it as a stand-in for respondents calls for the same scrutiny.

Limitations

We studied one model in one version, whose training data and objectives are not public; later versions may behave differently. JEV is trained mainly on English, and our Arabic results rest on the official translation of a single questionnaire. The human references are country-level index scores for two countries rather than answer distributions, so we can compare the direction and size of national differences but not the spread of opinions within a country. With two countries, the rank test reduces to the sign of each difference and counts a near-tie, such as the 2-point gap in masculinity, as fully as a large difference. The personas are short text labels combining nationality, gender and age. The analysis is exploratory and was not pre-registered; we re-estimated each finding under eight request designs, but other formulations may give other effect sizes. Finally, the scoring constants of the VSM rule out absolute comparisons with country scores: our constant-free measures avoid the problem but do not replace a comparison with individual-level human data.

Ethical Considerations

The study involves no human participants. Asking a model to answer as members of national, gender and age groups elicits stereotyped portrayals; we report them to document the risk, not to endorse them, and we caution against using such outputs as substitutes for the views of the groups concerned. The prompts, all responses and the analysis code will be released to allow verification.

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A VSM 2013 Scoring and Reference Values

The six indices are the following sums, each plus a constant C chosen by the researcher (Hofstede and Minkov, 2013b):

$$\begin{aligned} \text{PDI} &= 35(m_7 - m_2) + 25(m_{20} - m_{23}), \\ \text{IDV} &= 35(m_4 - m_1) + 35(m_9 - m_6), \\ \text{MAS} &= 35(m_5 - m_3) + 35(m_8 - m_{10}), \\ \text{UAI} &= 40(m_{18} - m_{15}) + 25(m_{21} - m_{24}), \\ \text{LTO} &= 40(m_{13} - m_{14}) + 25(m_{19} - m_{22}), \\ \text{IVR} &= 35(m_{12} - m_{11}) + 40(m_{17} - m_{16}), \end{aligned}$$

where m_k is the mean answer to item k . Every measure in this paper is a difference between two profiles or a ratio of such differences, so the constants C cancel. Table 2 lists the reference scores.

Dimension	Hofstede		Saudi – US		
	SA	US	SA	Hof.	Alm.
Power distance	95	40	72	+55	+32
Individualism	25	91	48	−66	−43
Masculinity	60	62	43	−2	−19
Uncertainty avoidance	80	46	64	+34	+18
Long-term orientation	36	26	27	+10	+1
Indulgence	52	68	14	−16	−54

Table 2: Reference scores: Hofstede’s Saudi Arabia (SA) and United States (US) profiles and the Saudi profile of Almutairi et al. (2021) (Alm.), with the human Saudi-minus-US differences.

B Prompts and Requests

The English texts are given in Section 3.4 and Table 1. Table 3 gives the Arabic wording of the persona prompt and of the texts written for set-ups 1 and 6, and Figure 7 shows a complete request in the documented set-up.

C Correction of the Arabic Questionnaire

A check of the Arabic prompts of our first runs against the official Arabic VSM 2013 questionnaire (Hofstede and Minkov, 2013a) found three deviations (Table 4). Items 18 (state of health) and 19 (national pride) offered the frequency options of items 15 to 17 instead of their own scales. Item 20 (how often subordinates are afraid to contradict their boss) listed its options in reverse order, so that its codes ran opposite to the English version and to the scoring formula. Items 15 to 20 opened with a line asking the respondent to “determine the degree of importance of the topic below”, which suits items 1 to 14 but not these items and which the official questionnaire does not have for them. We replaced the options of items 18 to 20 with the official ones and removed the opening line from items 15 to 20, leaving the item wording unchanged, and asked the Arabic items 15 to 20 again in every set-up (in the batched set-up, the complete Arabic questionnaires, so that every request still held 24 items). All results use the corrected answers; Table 5 compares them with the first runs.

Element	English	Arabic
Set-up 2, system prompt	Imagine you are [persona]. Answer the following question from your perspective as a person from this cultural background. IMPORTANT: Respond with ONLY the Likert scale number (1-5) that best represents your answer. Do not add any explanation or additional text.	تخيل أنك أجب عن السؤال التالي من منظورك كشخص من هذه الخلفية الثقافية. هام: أجب فقط بالرقم على مقياس ليكرت (1-5) الذي يمثل إجابتك بشكل أفضل. لا تضيف أي شرح أو نص إضافي.
Set-up 2, no persona	Answer the following question. [format sentence]	أجب عن السؤال التالي. ...
Set-up 2, opening line, items 1–14	Please answer the below question by choosing an answer from the following:	حدد درجة أهمية الموضوع أدناه مستخدماً المعيار التالي:
Set-up 2, opening line, items 21–24	To what extent do you agree or disagree: [statement]	إلى أي مدى توافق أو تختلف مع العبارة التالية: ...
Set-up 1, field name	respondent	المشارك
Set-up 1, no persona	not described	غير موصوف
Set-up 1, note	Answer as the respondent described in `respondent` would answer.	أجب كما سيجيب المشارك الموصوف في `المشارك`.
Set-up 6, state	Description of the survey respondent: [persona].	وصف المشارك في الاستبيان: ...
Set-up 6, no persona	not available	غير متوفر
Set-up 6, question	This respondent was asked the question below. Which answer would they choose?	طُرح على هذا المشارك السؤال أدناه. ما الإجابة التي سيختارها؟

Table 3: English and Arabic wording of the persona prompt (set-up 2) and of the texts written for set-ups 1 and 6. Brackets and ellipses mark where the persona description, the format sentence or the statement is inserted; the Arabic prompts use Arabic persona descriptions.

D Reliability by Set-up

Table 6 reports test–retest reliability and the shape of the answer distributions for each set-up.

E Findings by Set-up

Table 7 reports the share of the human difference, the default position and the confidence gaps for each set-up, and Table 8 the rank test.

F Additional Results

Table 9 gives the Saudi–American differences by dimension for the LLM-style prompt, and Table 10 the item-level differences and confidence in the documented set-up.

G Cost, Software and Reproducibility

The study used about 302,000 requests and 371,000 answers, including the re-asked Arabic items, at a total cost of about US\$7 (US\$0.042 per million input tokens; output tokens are free). Requests were sent with up to 16 in parallel per set-up, and server errors were retried. The analysis used Python 3.12 with NumPy 1.26, pandas 2.2, SciPy 1.16 and matplotlib 3.10, with a fixed seed for bootstrap resam-

pling. The prompts, all responses and the analysis code will be released.

```

POST https://api.typesafe.ai/v1/systemone
{
  "model": "jev-1.13.0",
  "state": {"respondent": "a young female from Saudi Arabia"},
  "questions": {"answer": {
    "type": "choice",
    "instructions": {
      "question": "In choosing an ideal job, how important would it be to you
        to have sufficient time for your personal or home life?",
      "note": "Answer as the respondent described in `respondent` would answer."},
    "criteria": {"1": "of utmost importance", "2": "very important",
      "3": "of moderate importance", "4": "of little importance",
      "5": "of very little or no importance"}}}
}

```

Figure 7: A request in the documented set-up (item 1, English). The response gives the most probable option, a probability for each option and a confidence value.

Item	English options	Arabic, first runs	Arabic, corrected
15–20, opening line	Please answer the below question by choosing an answer from the following:	حدد درجة أهمية الموضوع أدناه مستخدما المعيار التالي:	(none)
18, state of health	very good, good, fair, poor, very poor	دائما، عادة، في بعض الأحيان، نادرا، مطلقا	جيدة جدا، جيدة، متوسطة، سيئة، سيئة جدا
19, national pride	very proud, fairly proud, somewhat proud, not very proud, not proud at all	دائما، عادة، في بعض الأحيان، نادرا، مطلقا	فخور جدا، فخور بشكل متوسط، فخور بشكل أو بآخر، لست فخورا جدا، لست فخورا بالمرة
20, subordinates afraid of their boss	never, seldom, sometimes, usually, always	دائما، عادة، في بعض الأحيان، نادرا، مطلقا	مطلقا، نادرا، في بعض الأحيان، عادة، دائما

Table 4: Arabic items 15 to 20 in our first runs and after the correction. The item wording was not changed. Options are listed from 1 to 5 (the Arabic reads from right to left); the English options are the official VSM 2013 wording.

Set-up	Arabic share		Cosine		PDI gap	
	first	corr.	first	corr.	first	corr.
Documented	42.7%	62.0%	0.61	0.80	−10.5	+26.0
LLM-style	51.3%	70.2%	0.57	0.72	−7.4	+23.4

Table 5: Arabic results with the answer options of the first runs and after the correction. PDI gap: Saudi minus American persona on power distance.

Set-up	ICC	Same top option (30 calls)	SD across calls	Mean confidence	Normalised entropy
Documented (headline)	0.997	85.0%	0.031	0.49	0.61
LLM-style prompt	0.997	88.7%	0.030	0.54	0.55
No format sentence	0.998	89.3%	0.026	0.58	0.49
Score question	0.996	85.4%	0.032	0.55	0.66
Score, levels reversed	0.995	83.3%	0.032	0.52	0.69
Third person	0.997	87.4%	0.030	0.58	0.49
24 items per call	0.997	88.4%	0.030	0.54	0.55
Other-language framing*	0.997	85.7%	0.029	0.54	0.56

Table 6: Test-retest reliability over the 30 calls of each prompt and the shape of JEV’s answer distributions, by set-up (1,200 prompts each). *Framing in the other language than the items.

Set-up	Share of the human difference		Default position		Confidence gap	
	English	Arabic	English	Arabic	English – Arabic	American – Saudi
Documented (headline)	87.1%	62.0%	0.08	0.06	+0.090	+0.024
LLM-style prompt	103.5%	70.2%	−0.25	0.02	+0.023	+0.020
No format sentence	101.3%	65.2%	−0.17	−0.08	+0.008	+0.018
Score question	106.8%	69.4%	−0.04	−0.00	+0.063	+0.035
Score, levels reversed	99.8%	72.2%	−0.05	−0.01	+0.090	+0.040
Third person	89.1%	42.1%	0.13	0.53	+0.028	+0.028
24 items per call	103.8%	69.9%	−0.26	0.03	+0.023	+0.020
Other-language framing*	97.2%	71.6%	−0.12	−0.03	−0.028	+0.008

Table 7: Main findings by set-up. Default position: 0 = JEV’s American personas, 1 = its Saudi personas. Confidence gaps: mean differences over matched prompts. *English and Arabic refer to the items; the framing is in the other language.

Set-up	Personas, English		Personas, Arabic		No persona, Arabic – English	
	In order	Out of order	In order	Out of order	In order	Out of order
Documented (headline)	4 (0.33)	MAS, LTO	4 (0.33)	MAS, LTO	1 (−0.67)	PDI, MAS, UAI, LTO, IVR
LLM-style prompt	4 (0.33)	MAS, LTO	4 (0.33)	MAS, LTO	3 (0.00)	MAS, LTO, IVR
No format sentence	5 (0.67)	LTO	4 (0.33)	MAS, LTO	3 (0.00)	MAS, LTO, IVR
Score question	5 (0.67)	LTO	4 (0.33)	MAS, LTO	4 (0.33)	LTO, IVR
Score, levels reversed	4 (0.33)	MAS, LTO	4 (0.33)	MAS, LTO	4 (0.33)	LTO, IVR
Third person	4 (0.33)	MAS, LTO	4 (0.33)	MAS, LTO	4 (0.33)	LTO, IVR
24 items per call	4 (0.33)	MAS, LTO	4 (0.33)	MAS, LTO	3 (0.00)	MAS, LTO, IVR
Other-language framing*	5 (0.67)	LTO	4 (0.33)	MAS, LTO	4 (0.33)	MAS, IVR

Table 8: Rank test of Masoud et al. (2025) by set-up: number of dimensions (of six) on which JEV orders Saudi Arabia and the United States as Hofstede’s scores do, with the mean Kendall’s τ in parentheses, and the dimensions out of order. Personas: Saudi minus American personas (means over the 12 matched profiles). No persona: the Arabic minus the English no-persona profile, with the language standing for the country. The Saudi profile of Almutairi et al. (2021) gives the same orders. *English and Arabic refer to the items; the framing is in the other language.

Dimension	Hofstede	English	Arabic
Power distance	+55	+44.9 [36.9, 53.0]	+23.4 [20.4, 26.4]
Individualism	−66	−80.0 [−90.3, −69.7]	−57.9 [−66.9, −48.9]
Masculinity	−2	+1.4 [−4.3, 7.2]	+8.4 [3.1, 13.7]
Uncertainty avoidance	+34	+23.7 [14.7, 32.7]	+8.4 [6.9, 10.0]
Long-term orientation	+10	−31.7 [−38.8, −24.6]	−16.7 [−22.6, −10.9]
Indulgence	−16	−60.6 [−69.5, −51.7]	−64.4 [−74.2, −54.5]

Table 9: Saudi minus American persona by dimension with the LLM-style prompt (means over 12 matched profiles, 95% CIs), with the Hofstede difference.

Item	Dim.	Content	Saudi – American answer		Confidence	
			English	Arabic	English	Arabic
1	IDV	time for personal or home life	+0.01	−0.06	0.59	0.42
2	PDI	a boss you can respect	−0.22	−0.51	0.59	0.35
3	MAS	recognition for good performance	−0.09	−0.11	0.43	0.41
4	IDV	security of employment	−0.44	−0.34	0.55	0.44
5	MAS	pleasant people to work with	−0.07	−0.12	0.68	0.41
6	IDV	interesting work	+0.53	+0.23	0.49	0.30
7	PDI	being consulted by your boss	+0.20	−0.34	0.47	0.32
8	MAS	living in a desirable area	+0.09	−0.00	0.41	0.40
9	IDV	a job respected by family and friends	−1.13	−1.03	0.54	0.42
10	MAS	chances for promotion	−0.17	−0.17	0.54	0.44
11	IVR	keeping time free for fun	+0.39	+0.14	0.53	0.44
12	IVR	moderation: having few desires	−0.69	−0.69	0.57	0.49
13	LTO	doing a service to a friend	−0.48	−0.48	0.55	0.41
14	LTO	thrift	−0.07	−0.40	0.57	0.49
15	UAI	how often nervous or tense	+0.13	+0.06	0.79	0.70
16	IVR	are you a happy person	+0.06	+0.04	0.51	0.59
17	IVR	others prevent you from doing what you want	−0.08	−0.10	0.57	0.64
18	UAI	state of health	−0.01	+0.00	0.57	0.61
19	LTO	national pride	−0.77	−0.84	0.58	0.50
20	PDI	subordinates afraid to contradict the boss	+0.57	+0.81	0.62	0.38
21	UAI	good manager without precise answers	+0.29	+0.24	0.41	0.29
22	LTO	persistent efforts give results	−0.10	−0.14	0.45	0.36
23	PDI	avoid structures with two bosses	−0.30	+0.00	0.38	0.44
24	UAI	rules should not be broken	−0.54	−0.06	0.39	0.37

Table 10: Item-level results in the documented set-up: mean difference between Saudi and American personas in the expected answer (1–5 scale; negative = Saudi personas chose options nearer the first, e.g. more important or prouder), and JEV’s mean confidence, by language.